Polarized neutron diffraction with *ex-situ* ^{3}He neutron spin filter and two-dimensional detector for a complex incommensurate magnetic order

Shingo Takahashi,[1,2] Yuta Osawa,[1] Ryuju Kobayashi,[2] Takashi Ino,[3] Hiraku Saito,[1] Takuro Kawasaki,[2] Takayuki Oku,[2] Tatsuya Nakamura,[2] Yoichi Ikeda,[4] Masaki Fujita,[4] Daisuke Koto,[5] Yusuke Tokunaga,[5] Taka-hisa Arima,[5,6] and Taro Nakajima[1,3,6]

[1]*Institute for Solid State Physics, The University of Tokyo, Kashiwa, Chiba 277-8581, Japan*
[2]*J-PARC Center, Japan Atomic Energy Agency, Tokai, Ibaraki 319-1195, Japan*
[3]*Institute of Materials Structure Science, High Energy Accelerator Research Organization, Tsukuba, Ibaraki 305-0801, Japan*
[4]*Institute for Materials Research, Tohoku University, Sendai, Miyagi 980-8577, Japan*
[5]*Department of Advanced Materials Science, The University of Tokyo, Kashiwa, Chiba 277-8561, Japan*
[6]*RIKEN Center for Emergent Matter Science, Wako, Saitama 351-0198, Japan*

Polarized neutron scattering has been widely used for studying magnetic orders in condensed matter. One of the typical setups for this technique is a triple-axis spectrometer with a ferromagnetic single-crystal monochromator and analyzer, which is suited for point-by-point measurements within the horizontal scattering plane. However, this setup cannot fully meet the demands of recent studies on complex magnetic orders with emergent cross-correlated phenomena. For instance, multiferroics, magnetic skyrmions, and other topologically nontrivial magnetic orders tend to have incommensurate magnetic modulation vectors running along various directions in reciprocal space, which cannot be captured by measurements restricted to the horizontal scattering plane. To overcome these limitations, we developed an experimental setup combining a ^{3}He neutron spin filter (^{3}He-NSF) and a two-dimensional (2D) detector, which enables us to analyze the polarization of neutrons scattered in the out-of-plane direction. We also established a data reduction procedure to convert the 2D intensity maps into three-dimensional intensity distributions in reciprocal space, correcting for the effect of imperfect ^{3}He polarization. This method was applied to single-crystal diffraction measurements on the multiferroic material Mn_3WO_6, which exhibits complex incommensurate magnetic orders. We successfully observed polarization-dependent intensities of out-of-plane incommensurate magnetic reflections, demonstrating the effectiveness of the present method for analyzing complex magnetic structures.

I. INTRODUCTION

Polarized neutron scattering is a powerful technique for investigating magnetic order and excitations in condensed matter. This capability arises from the magnetic moment of the neutron, which directly interacts with magnetic moments in the material through the magnetic dipole-dipole interaction. By analyzing the spin states of neutrons before and after scattering, one can separate magnetic scattering from nuclear scattering and determine the directions of magnetic moments in the sample. To perform these measurements, devices for controlling and probing the neutron polarization are required. To date, three techniques have been widely used for polarized neutron scattering. The first is a single crystal of ferromagnetic material, such as a Cu_2MnAl Heusler compound. By exploiting the nuclear-magnetic interference effect, which gives rise to different Bragg reflectivities for the two neutron spin states, a polarized monochromatic neutron beam can be obtained [1]. The second is a polarizing supermirror, which utilizes the difference in critical angle between neutrons with different spin states [2]. The third is a polarizing filter, utilizing the strong spin dependence of the neutron absorption cross section of ^{3}He nuclei (^{3}He neutron spin filter, ^{3}He-NSF) [3]. Among these techniques, the ferromagnetic single-crystal monochromators are often used to polarize incident neutrons and analyze scattered neutrons in triple-axis spectrometers [4–8], as shown in Fig. 1(a). While this type of instrument has been widely used for studying magnetic structures, accessible reflections are often restricted to the horizontal scattering plane.

In recent years, increasing attention has been paid to materials hosting complex incommensurate magnetic orders, such as spin-driven multiferroics, magnetic skyrmions and related topological magnetic orders [9–13]. In these systems, the magnetic moments are often arranged in noncollinear or noncoplanar configurations, which can give rise to emergent cross-correlated phenomena. To study these systems, it is necessary to measure magnetic satellite reflections corresponding to magnetic modulation wavevectors (q-vectors), which often run along various directions in reciprocal space. Consequently, three-dimensional exploration of the reciprocal space with a two-dimensional (2D) detector is of great importance. To realize polarization analysis in the measurements with a 2D detector, polarization devices with a large solid-angle coverage, such as ^{3}He-NSF, are required.

The combination of a ^{3}He-NSF and a 2D detector has been implemented in several small-angle neutron scattering (SANS) instruments and has been applied to studies of complex magnetic structures [14, 15]. However, their accessible reciprocal space region is generally limited to relatively low-Q. To extend the accessible Q range for polarization analysis to middle-Q and high-Q regions, it would be necessary to place a ^{3}He-NSF and a 2D detector at finite scattering angles, as illustrated in Fig. 1(b). Related instrumental concepts have already been reported at several neutron facilities, including POLI at MLZ [16] and WOMBAT at ANSTO [17]. Nevertheless, to the best of our knowledge, quantitative experimental demonstrations of polarization analysis using this type of setup remain limited. Other method to perform polarization analysis from low-Q to high-Q range is using a wide-angle ^{3}He cell. This type of ^{3}He cell are used in several instruments, such as MACS at NIST [18–20], D1B at ILL [21], and LET at ISIS [22, 23]. However, these instruments are not equipped with a 2D detector, and thus the polarization analysis is limited to the horizontal scattering plane.

In the present study, we performed polarized neutron diffraction measurements combining a ^{3}He-NSF and a 2D detector on the 5G beamport at JRR-3. This paper mainly describes the experimental setup, the measurement procedures, and corrections for the effects of imperfect neutron polarization. We also performed polarized neutron diffraction on a single crystal of Mn_3WO_6, which exhibits incommensurate magnetic orders. This compound has a crystal structure belonging to the space group $R\bar{3}$ and displays two magnetically ordered phases at low temperatures [24]. As we describe in the following sections, we focus on the data measured at 23 K, which corresponds to the high-temperature magnetic phase. We performed polarization analysis for magnetic reflections that were slightly shifted from the horizontal plane. The details of the magnetic structure will be reported elsewhere [25].

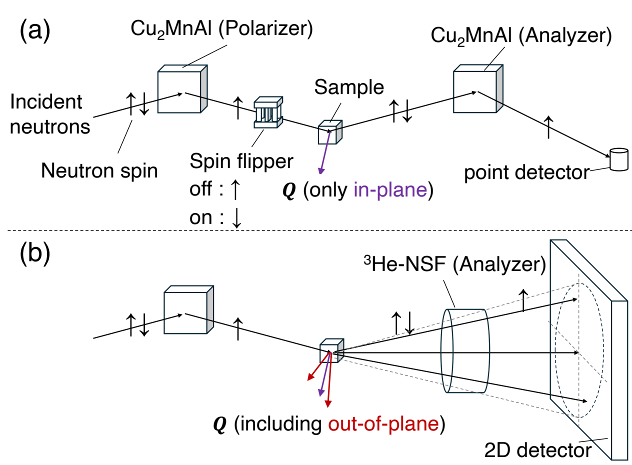

FIG. 1. Schematic illustrations comparing neutron polarization analysis techniques. (a) Conventional setup using Heusler crystals for both the polarizer and the analyzer, where only in-plane scattered neutrons are detected by a point detector. (b) Polarization analysis setup combining a ^{3}He-NSF and a 2D detector, enabling detection of both in-plane and out-of-plane scattered neutrons.

II. EXPERIMENTAL DETAILS

A. Setup

The 5G beamport at JRR-3 is normally operated as the POlarized Neutron Triple-Axis spectrometer (PONTA), which has a longitudinal polarization analysis option with a conventional Heusler monochromator and analyzer [8]. In the present experiment, the Heusler analyzer and the point detector were replaced with a ^{3}He-NSF and a 2D detector. The Heusler monochromator provided a polarized incident neutron beam with an energy of 14.7 meV. The higher-order reflections from the monochromator were suppressed by a pyrolytic graphite filter placed between the monochromator and the sample. Figure 2 shows an overview of the experimental setup from the sample to the detector. Scattered neutrons were detected using a scintillation-type 2D detector employing wavelength-shifting fiber readout [26, 27]. The active area is 256×256 mm^2, consisting of 64×64 channels with a channel pitch of 4 mm. The detector was positioned at an angle of 32° with respect to the incident neutron beam and remained fixed throughout the measurement. The sample-to-detector distance was 915 mm. A ^{3}He-NSF was installed 430 mm downstream of the sample. The cylindrical ^{3}He cell provided an angular acceptance of $\pm 4.0°$, which corresponded to a circular region with a radius of 76 mm (approximately 19 channels) centered on the detector. Polarization analysis of the scattered neutrons was possible only within this region. To reduce the background, a B_4C-shielded neutron guide tube was installed between the ^{3}He-NSF and the detector, and B_4C shielding was also attached around the ^{3}He-NSF coil. The non-spin-flip and spin-flip intensities of the scattered neutrons were obtained by flipping the ^{3}He polarization. Therefore, the neutron spin flipper for the incident beam was not used. A single-crystal sample of Mn_3WO_6 with a volume of 8 mm^3 was mounted on an aluminum holder and installed in a ^{4}He closed-cycle refrigerator such that the $(H, 0, L)$ plane coincided with the horizontal scattering plane. The sample rotation angle ω was adjustable to align the target reflection with the detector. The Helmholtz coil was used to maintain the neutron spin polarization along the neutron flight path by applying a vertical magnetic field. The magnetic field in the ^{3}He-NSF was oriented parallel to the beam axis, and the spins of neutrons scattered from the sample adiabatically followed the field direction, as illustrated in Fig. 2. The magnetic field strength generated by both the ^{3}He-NSF coil and the Helmholtz coil was approximately 1 mT.

B. ^{3}He neutron spin filter

In this section, we describe the ^{3}He-NSF employed in the present experiment in detail. Neutrons with spins antiparallel to the ^{3}He nuclear spins are strongly ab-

TABLE I. Specifications of ^{3}He cells.

Name	$D \times L$ (mm)	Volume (cm^3)	^{3}He density (amg)	N_2 pressure (Torr)	K/Rb (mol)
Kenji	72 × 67	140	3.2	97	26
Shimpachi	72 × 65	146	3.1	100	37

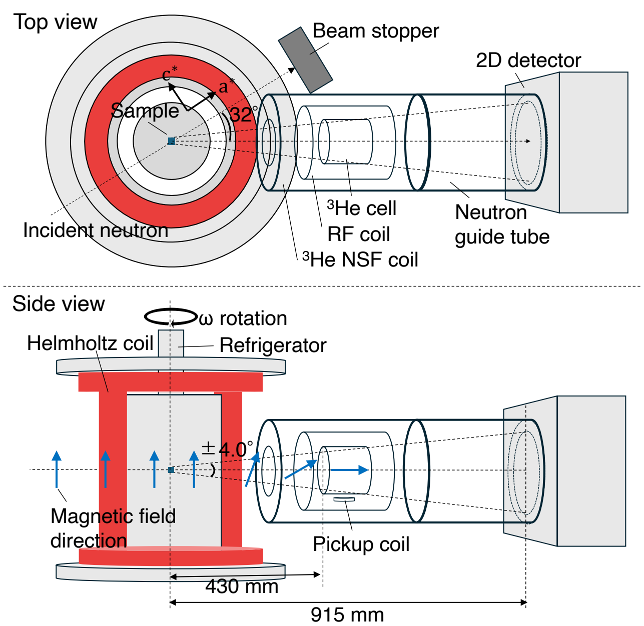

FIG. 2. Schematic illustration of the experimental setup on the 5G beamport. Scattered neutrons from the sample pass through the ^{3}He-NSF coil and the neutron guide tube before reaching the 2D detector. The sample is mounted in a cryostat whose rotation angle is denoted by ω. The magnetic field direction induced by the Helmholtz coil is vertical, whereas the field within the ^{3}He-NSF is oriented along the beam axis. The neutron spins are adiabatically rotated to follow the field direction. A radio-frequency (RF) coil is used to flip the ^{3}He polarization, and an NMR pickup coil monitors the ^{3}He polarization after each spin flip.

sorbed, while those with spins parallel to the ^{3}He nuclear spins are transmitted. Thus, neutrons are polarized by passing through the polarized ^{3}He gas, and neutron polarization and transmission depend on the ^{3}He polarization. The two major methods for polarizing ^{3}He nuclei are spin-exchange optical pumping (SEOP) [28] and metastability-exchange optical pumping (MEOP) [29]. In the present study, we employed the SEOP method. In the SEOP method, a glass cell is filled with ^{3}He , N_2, and a small amount of alkali metal. The cell is irradiated with circularly polarized light resonant with an optical transition of the alkali metal. The alkali-metal electrons undergo repeated excitation and de-excitation, with N_2 serving as a quenching gas to suppress radiative decay, resulting in electron spin polarization. The polarization of the alkali metal is then transferred to the ^{3}He nuclei via spin-exchange collisions, thereby achieving ^{3}He nuclear polarization. The SEOP system used in this study was based on the system developed for the polarized neutron spectrometer POLANO, installed at BL23 in MLF of J-PARC [30]. Both the ^{3}He-NSF coil used for polarization analysis shown in Fig. 2 and the SEOP system are equipped with an adiabatic fast-passage nuclear magnetic resonance (AFP-NMR) system [31]. The AFP-NMR system consists of a radio-frequency (RF) coil for flipping the ^{3}He polarization and a pickup coil located at the bottom of the ^{3}He cell for monitoring the ^{3}He polarization. Two cylindrical ^{3}He glass cells, named Kenji and Shimpachi, made of aluminosilicate glass (GE180) with nearly identical dimensions, were used in this experiment. Both cells were filled with ^{3}He gas, nitrogen gas, and a mixture of alkali-metal atoms (Rb and K) for hybrid SEOP, which provides a higher pumping efficiency than Rb-only SEOP [32, 33]. The details of the cell specifications are described in Table I. A darkroom to house the SEOP laser system, where the ^{3}He gas was polarized, was constructed in the 5G beamport experimental area. After the ^{3}He polarization reached saturation, only the ^{3}He cell was transported to the ^{3}He-NSF coil on the beamline. Neutron measurements were performed during the relaxation of the ^{3}He polarization, with the neutron polarization and transmission decreasing. This technique is known as the *ex-situ* ^{3}He-NSF method. Since the ^{3}He polarization decreases with time in this method, two cells were used alternately for the experiment. While one cell was used for the polarized neutron experiment, the other was polarized in the SEOP system, and the cells were exchanged every two or three days.

III. DATA REDUCTION PROCEDURE INCLUDING POLARIZATION CORRECTION

Figure 3 shows the procedure for data acquisition, reduction, and polarization correction in the present study. At a sample rotation angle of $\omega = \omega_n$, we measured a pair of 2D intensity maps corresponding to non-spin-flip and spin-flip scattering by flipping the ^{3}He polarization. We refer to the observed intensities measured with the ^{3}He nuclear polarization parallel or antiparallel to the neutron polarization produced by the Heusler polarizer as $I_{\rm obs}^{++}$ and $I_{\rm obs}^{+-}$, respectively. Note that these intensity maps were normalized by the measurement time. The measurement time for each intensity map (20 min) was sufficiently shorter than the relaxation time of the ^{3}He polarization, and thus the transmission and polarization produced by the ^{3}He-NSF can be approximated by their values at the midpoint of each measurement. As shown

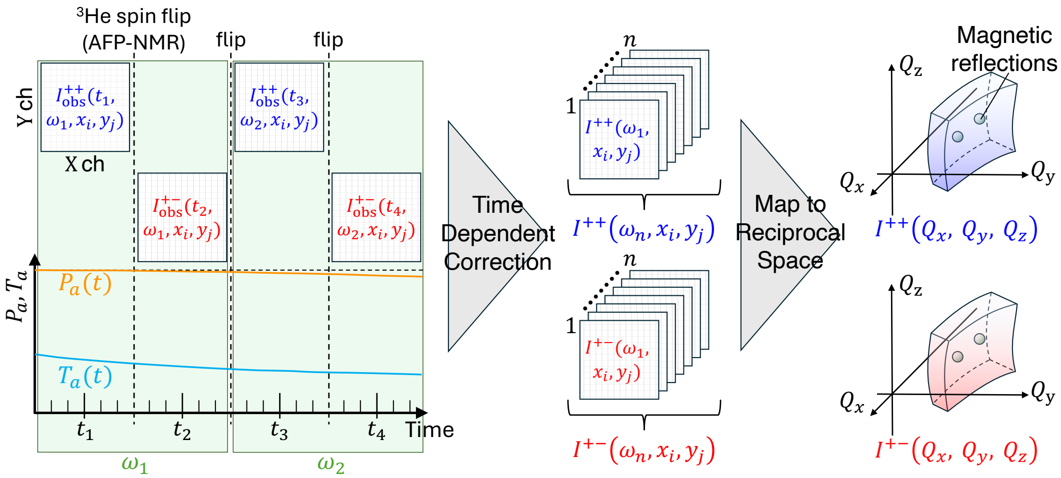

FIG. 3. Schematic diagram of the data-reduction procedure for polarization analysis using an *ex-situ* ^{3}He-NSF and a 2D detector. Raw 2D detector data measured at different sample rotation angles ω and times t are corrected for the time-dependent neutron polarization $P_a(t)$ and transmission $T_a(t)$ of the ^{3}He-NSF. The detector coordinates (X,Y) and sample rotation angle ω are converted to Q_x, Q_y, Q_z, and then transformed to H, K, L using the UB matrix. The corrected three-dimensional reciprocal-space data can then be sliced along an arbitrary plane.

in left panel of Fig. 3, we define the midpoint times of the two successive measurements as t_{2n-1} and t_{2n}, respectively. Using this notation, the intensities observed at the pixel (x_i, y_j) are described as

$$\begin{aligned} I^{++}_{\rm obs}(\omega_n, t_{2n-1}, x_i, y_j) = T_p T_a(t_{2n-1}) \Big[&(1 + P_p P_a(t_{2n-1}))\, I^{++}(\omega_n, x_i, y_j) \\ &+ (1 - P_p P_a(t_{2n-1}))\, I^{+-}(\omega_n, x_i, y_j) \Big] + \mathrm{BG}(x_i, y_j), \end{aligned} \tag{1}$$

$$\begin{aligned} I^{+-}_{\rm obs}(\omega_n, t_{2n}, x_i, y_j) = T_p T_a(t_{2n}) \Big[&(1 - P_p P_a(t_{2n}))\, I^{++}(\omega_n, x_i, y_j) \\ &+ (1 + P_p P_a(t_{2n}))\, I^{+-}(\omega_n, x_i, y_j) \Big] + \mathrm{BG}(x_i, y_j), \end{aligned} \tag{2}$$

where $I^{++}(\omega_n, x_i, y_j)$ and $I^{+-}(\omega_n, x_i, y_j)$ are the ideal non-spin-flip and spin-flip scattering intensities, respectively. T_p and $T_a(t)$ denote the neutron transmissions of the Heusler polarizer and ^{3}He-NSF, respectively. P_p and $P_a(t)$ are the neutron polarizations produced by the Heusler polarizer and ^{3}He-NSF, respectively. In general, a full polarization analysis experiment, one would measure four scattering intensities, I^{++}, I^{+-}, I^{-+}, and I^{--}, corresponding to the four possible combinations of the polarizer and analyzer spin state [34]. These comprise two non-spin-flip channels, I^{++} and I^{--}, and two spin-flip channels, I^{+-} and I^{-+}. From these four scattering intensities, the proportions of nuclear, magnetic and incoherent scattering can be determined. However, in the present case, we assume that the sample is an antiferromagnet in which nuclear and magnetic reflections are separated from each other. Thus, we can impose the conditions of $I^{++} = I^{--}$ and $I^{+-} = I^{-+}$, and can simplify the equations regarding the effect of imperfect polarization described in Appendix A. For the Heusler polarizer, transmission corresponds to the ratio of the Bragg-reflected intensity to the incident unpolarized neutron intensity. We also note that the chiral magnetic scattering, which depends on the sign of the incident neutron polarization direction, is negligible because the neutron polarization and the scattering vector are nearly perpendicular to each other [35]. Note that they are defined for an incident unpolarized neutron beam. The values of T_p and P_p for the Heusler polarizer are treated as time-independent, whereas $T_a(t)$ and $P_a(t)$ for the ^{3}He-NSF are time-dependent. $\mathrm{BG}(x_i, y_j)$ is the background intensity at the detector pixel (x_i, y_j). In the present experiment, the background signals were mainly due to imperfect shielding of the detector and the neutron flight path. Thus, we assumed that the background signals are independent of time, ^{3}He polarization, and the sample rotation angle ω. To measure the background data, we set the ω angle in order to avoid measuring any Bragg reflections from the sample, and obtained $\mathrm{BG}(x_i, y_j)$. We repeated the measurements of I^{++}_{obs} and I^{+-}_{obs} while varying the sample rotation angle ω.

By solving Eqs. 1 and 2 for I^{++} and I^{+-}, we obtain the equations of the polarization correction for the dataset measured at $\omega = \omega_n$ as follows:

$$
\begin{aligned}
I^{++}(\omega_n, x_i, y_j) =& \frac{1 + P_p P_a(t_{2n})}{2T_p T_a(t_{2n-1}) P_p \left[P_a(t_{2n-1}) + P_a(t_{2n})\right]} \left[I^{++}_{\mathrm{obs}}(\omega_n, t_{2n-1}, x_i, y_j) - \mathrm{BG}(x_i, y_j)\right] \\
&- \frac{1 - P_p P_a(t_{2n-1})}{2T_p T_a(t_{2n}) P_p \left[P_a(t_{2n-1}) + P_a(t_{2n})\right]} \left[I^{+-}_{\mathrm{obs}}(\omega_n, t_{2n}, x_i, y_j) - \mathrm{BG}(x_i, y_j)\right], \qquad (3)
\end{aligned}
$$

$$
\begin{aligned}
I^{+-}(\omega_n, x_i, y_j) =& - \frac{1 - P_p P_a(t_{2n})}{2T_p T_a(t_{2n-1}) P_p \left[P_a(t_{2n-1}) + P_a(t_{2n})\right]} \left[I^{++}_{\mathrm{obs}}(\omega_n, t_{2n-1}, x_i, y_j) - \mathrm{BG}(x_i, y_j)\right] \\
&+ \frac{1 + P_p P_a(t_{2n-1})}{2T_p T_a(t_{2n}) P_p \left[P_a(t_{2n-1}) + P_a(t_{2n})\right]} \left[I^{+-}_{\mathrm{obs}}(\omega_n, t_{2n}, x_i, y_j) - \mathrm{BG}(x_i, y_j)\right]. \qquad (4)
\end{aligned}
$$

As we show later, we also experimentally determined the neutron polarizations produced by the polarizer and the analyzer, P_p and $P_a(t)$, and the transmission of the analyzer, $T_a(t)$, including the temporal decay of the ^{3}He polarization. Since T_p is time-independent, it was instead treated as a constant scale factor. Substituting all the parameters into the equations above, we corrected the effect of imperfect polarization for all the pixels.

Finally, we converted the detector coordinates and sample rotation angle (x_i, y_j, ω_n) into reciprocal-space coordinates (Q_x, Q_y, Q_z). The resulting data were further transformed into (H, K, L) coordinates using the UB matrix, allowing the intensity distribution to be sliced along an arbitrary plane in reciprocal space.

IV. RESULTS

A. Evaluation of ^{3}He neutron spin filter from nuclear scattering

The ^{3}He polarization was evaluated by measuring a nuclear Bragg peak. Specifically, we selected the $\bar{1}02$ reflection, which has a similar scattering angle to that of the magnetic reflections of interest. The neutron transmission for unpolarized neutrons $T_a(t)$, the neutron polarization $P_a(t)$, and the ^{3}He polarization $P_{\mathrm{He}}(t)$ are given by the following equations [36]:

$$T_a(t) = T_{\mathrm{cell}} \exp(-\mathcal{O}) \cosh(\mathcal{O} P_{\mathrm{He}}(t)), \qquad (5)$$
$$P_a(t) = \tanh(\mathcal{O} P_{\mathrm{He}}(t)), \qquad (6)$$
$$P_{\mathrm{He}}(t) = P_{\mathrm{He}}(0) \exp(-t/\tau). \qquad (7)$$

Here, T_{cell} is the transmission of the glass walls of the ^{3}He cell. $\mathcal{O}$ is the opacity of the ^{3}He cell, defined as $\mathcal{O} = \rho d \sigma$. ρ is the number density of ^{3}He , and d is the neutron path length through the ^{3}He cell, which depends on the positions of the detector pixels. σ is the neutron absorption cross section of unpolarized ^{3}He , which depends on the incident neutron energy. According to the literature, σ for the incident neutron energy of 14.7 meV is 6997 barn [37]. $P_{\mathrm{He}}(0)$ is the initial polarization at $t = 0$, and τ is the relaxation time.

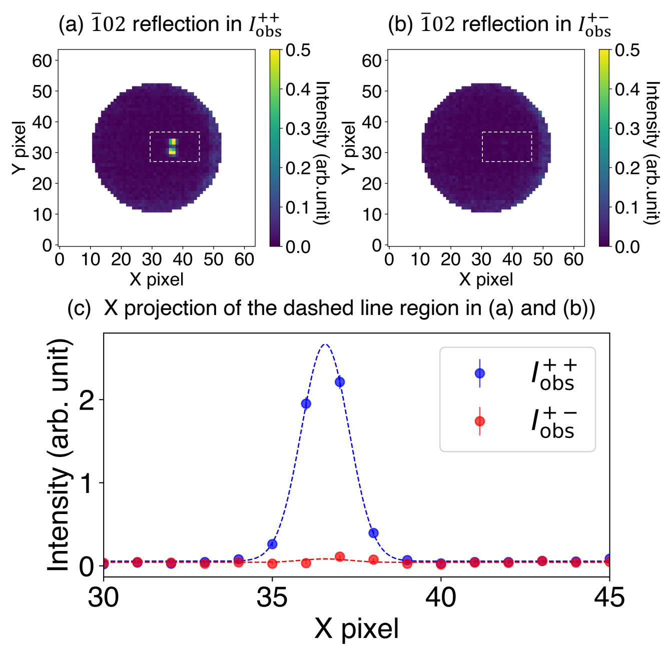

FIG. 4. (a) Observed non-spin-flip (I^{++}_{obs}) and (b) spin-flip (I^{+-}_{obs}) intensity maps, showing only the analyzing region. (c) Line profile along the X direction, obtained by integrating the intensity within the rectangular regions indicated by the dashed lines in (a) and (b).

First, the areal density ρd of each cell was determined as follows. Before polarizing ^{3}He by SEOP, we measured the nuclear peak through the unpolarized ^{3}He cell with ω fixed at the Bragg condition. From the intensity map,

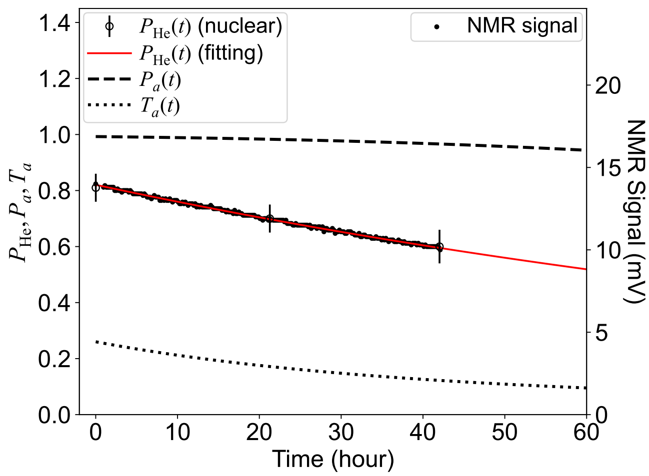

FIG. 5. Time dependence of the ^{3}He polarization, neutron polarization, and neutron transmission. The open circles show the ^{3}He polarization determined from the intensity of the $\bar{1}02$ nuclear reflection, while the closed circles show the NMR signal. The NMR signal was normalized to the initial ^{3}He polarization determined from the reflection. The red line represents the fitting result using Eq. 7. The dashed and dotted lines indicate the calculations for $P_a(t)$ and $T_a(t)$, respectively, obtained from $P_{\rm He}(t)$.

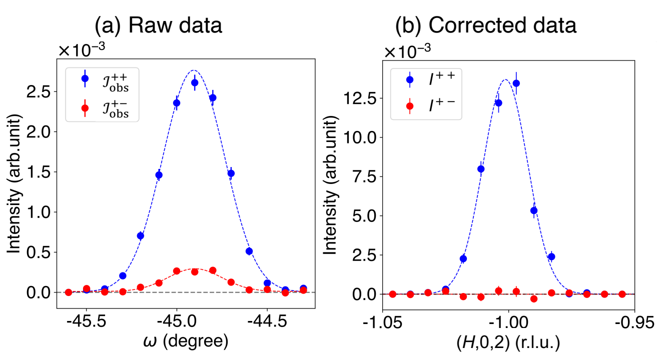

FIG. 6. (a) Raw non-spin-flip and spin-flip intensities, $\mathcal{I}^{++}_{\rm obs}$ (red) and $\mathcal{I}^{+-}_{\rm obs}$ (blue), obtained by integrating the peak intensity on the 2D detector while scanning the sample rotation angle ω. (b) Corrected non-spin-flip and spin-flip scattering intensities, $I^{++}(H,K,L)$ (red) and $I^{+-}(H,K,L)$ (blue) in reciprocal space. The data were obtained by slicing along the $(H,0,2)$ line in the three-dimensional reciprocal space and integrating over a thickness of $\pm$ 0.025 r.l.u. along the $(K/2,-K,0)$ direction, which is perpendicular to the horizontal $(H0L)$ scattering plane, and over $\pm$ 0.03 r.l.u. along the L direction.

we extracted a box-integrated intensity near the peak. After subtracting the background signals, we obtained the integrated intensity of the Bragg reflection, $\mathcal{I}^{\rm unpol}_{\rm obs}$. By using Eqs. 1, 5, and ^{3}He polarization is zero ($P_{\rm He}(t) = 0$), $\mathcal{I}^{\rm unpol}_{\rm obs}$ is expressed as follows:

$$\mathcal{I}^{\rm unpol}_{\rm obs} = \sum_{(i,j)\in R} T_p T_{\rm cell} \exp(-\mathcal{O}) I^{++}(x_i, y_j). \quad (8)$$

where R denotes the box-integrated region. Note that the nuclear Bragg reflection contains only the non-spin-flip scattering intensity, and thus we can impose $I^{+-} = 0$ in Eq. 1. The same measurement was performed using an empty cell with the same dimensions as the ^{3}He glass cell, yielding $\mathcal{I}^{\rm empty}_{\rm obs}$. Here, the transmission term $T_{\rm cell}\exp(-\mathcal{O})$ is replaced by $T_{\rm cell}$. Then, taking the ratio $\mathcal{I}^{\rm unpol}_{\rm obs}/\mathcal{I}^{\rm empty}_{\rm obs}$, the ρd values of the Kenji and Shimpachi cells were determined to be 18.5 ± 0.5 amg cm and 18.8 ± 0.4 amg cm, respectively. Note that the amagat is a non-SI unit of the volumetric number density of an ideal gas at 0 °C (273.15 K) and 1 atm (101.325 kPa), and one amagat is equivalent to $2.69 \times 10^{19}/\text{cm}^3$. The ρd values determined here correspond to the path lengths through the ^{3}He gas at the pixel positions of the detected nuclear peaks, and the path length d varies with pixel position. However, the solid angle covered by the cell was sufficiently small (up to 4°) that the variation in d could be neglected. Therefore, these ρd values were used for all pixels included in the present data analysis. $T_{\rm cell}$ was determined to be 0.91 ± 0.04 by comparing the nuclear peak intensities measured without the cell with the empty cell. This value was not directly used in the polarization analysis.

Second, we determined $P_{\rm He}(0)$ in Eq. 7 by measuring the nuclear reflection with the polarized ^{3}He cell. Immediately after moving the ^{3}He cell from the SEOP system to the beamline, we measured the intensities of the reflection with the ^{3}He polarization parallel and antiparallel to the incident neutron polarization. The integrated intensities were obtained following the same procedure as that used for the measurement with the unpolarized ^{3}He cell. These are referred to as $\mathcal{I}^{++}_{\rm obs}(t=0)$ and $\mathcal{I}^{+-}_{\rm obs}(t=0)$, respectively. Since these measurement times were much shorter than the ^{3}He relaxation time, the ^{3}He polarizations for the two measurements were assumed to be equal. As shown in Fig. 4, we clearly observed the nuclear Bragg peak in the non-spin-flip channel, while the intensity in the spin-flip channel was extremely weak. Note that the peak in Fig. 4(a) shows a slight splitting due to the mosaicity of the single-crystal sample. From Eqs. 1, 2, 5, and 8, the ^{3}He polarization $P_{\rm He}(t)$ at $t = 0$ satisfies the following relationship:

$$\frac{\mathcal{I}^{++}_{\rm obs}(t=0) + \mathcal{I}^{+-}_{\rm obs}(t=0)}{\mathcal{I}^{\rm unpol}_{\rm obs}} = 2\cosh(\mathcal{O}P_{\rm He}(0)). \quad (9)$$

We measured the nuclear Bragg reflection to estimate $P_{\rm He}(0)$ each time the cell was changed. For instance, the initial ^{3}He polarization of Shimpachi cell was determined to be $P_{\rm He}(0) = 0.81 \pm 0.05$.

We then measured the relaxation of the ^{3}He polarization by tracking the NMR signal of the ^{3}He-NSF, which is proportional to the ^{3}He polarization. Throughout the

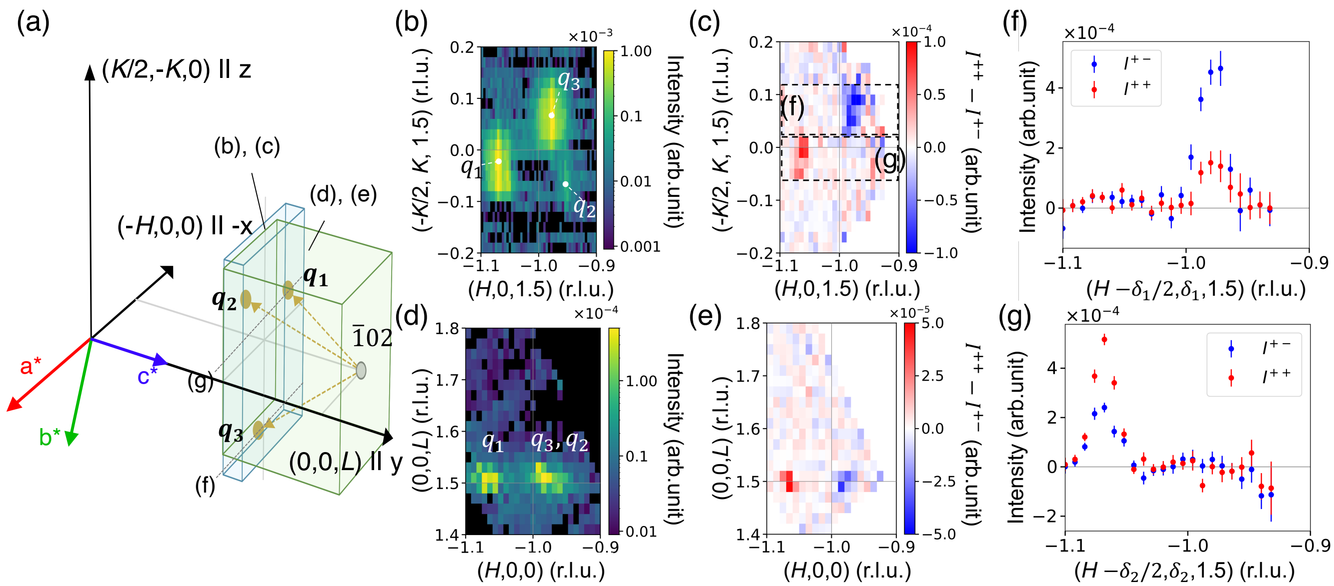

FIG. 7. (a) Schematic illustration of the reciprocal-space region measured in the present experiment. The translucent volumes labeled (b), (c) and (d), (e) indicate the regions integrated along the L direction and the $(-K/2, K, 0)$ direction, respectively. The dotted lines along the $-H$ direction through $\boldsymbol{q}_3$ and $\boldsymbol{q}_1$ indicate the line cuts shown in panels (f) and (g), respectively. (b), (d) Unpolarized neutron intensity maps in the $(H, K, 1.5)$ and $(H, 0, L)$ planes, respectively. (c), (e) Polarization-analysis maps obtained from the difference between the corrected non-spin-flip and corrected spin-flip intensities, $I^{++} - I^{+-}$. The intensity maps in the $(H, K, 1.5)$ plane shown in (b) and (c) were obtained from the three-dimensional reciprocal space data by integrating over $1.47 \leq L \leq 1.52$ r.l.u. Similarly, the intensity maps in the $(H, 0, L)$ plane shown in (d) and (e) were obtained by integrating along the $(-1/2, 1, 0)$ direction over the range of ± 0.1 r.l.u. (f), (g) Line profiles of I^{++} and I^{+-} through $\boldsymbol{q}_3$ and $\boldsymbol{q}_1$, respectively, obtained by projecting the intensities within the dotted boxes in (c) onto the horizontal axis. Here, the coordinate along the $(-1/2, 1, 0)$ direction is parameterized by δ, such that the horizontal axis is given by $(H - \delta/2, \delta, 1.5)$. The projected ranges were $0.015 \leq \delta_1 \leq 0.095$ for (f) and $-0.07 \leq \delta_2 \leq 0.02$ for (g).

neutron diffraction measurement, the NMR signal was recorded each time the ^{3}He polarization was flipped. Figure 5 shows the time dependence of the NMR signal of Shimpachi cell. By normalizing the NMR data to the $P_{\mathrm{He}}(0)$ determined by the nuclear reflection, we obtained the ^{3}He polarization as a function of time. We fitted Eq. 7 to the normalized NMR data and obtained P_{He} as shown by the red solid curve in Fig. 5. The relaxation times τ for Shimpachi and Kenji cells were determined to be $\tau = 131.1 \pm 0.7$ and $\tau = 128.4 \pm 0.3$ hours, respectively. We also measured the nuclear Bragg reflection several times during the experiment, confirming that the values of the ^{3}He polarization deduced from the measurements of the nuclear reflection and the NMR signal showed good agreement. The time evolutions of $T_a(t)$ and $P_a(t)$ were also obtained using Eqs. 5, 6, and 7, as shown in Fig. 5.

The neutron polarization produced by the Heusler polarizer, P_p, was also determined from the integrated intensities of the nuclear reflection. From Eqs. 1 and 2, the asymmetry between $\mathcal{I}^{++}_{\mathrm{obs}}$ and $\mathcal{I}^{+-}_{\mathrm{obs}}$ after subtracting the background $\mathrm{BG}(x_i, y_j)$ from each is given by

$$\frac{\mathcal{I}^{++}_{\mathrm{obs}} - \mathcal{I}^{+-}_{\mathrm{obs}}}{\mathcal{I}^{++}_{\mathrm{obs}} + \mathcal{I}^{+-}_{\mathrm{obs}}} = P_p P_a(t). \tag{10}$$

Since $P_a(t)$ has already been determined from the ^{3}He polarization, P_p can be obtained from this equation. The values of P_p deduced from the measurements of the nuclear reflection were averaged, yielding $P_p = 0.86 \pm 0.06$. Note that P_p includes the effects of environmental background and neutron depolarization in the guide field.

To check the validity of the polarization correction, we measured the intensity distributions near the $\bar{1}02$ reflection while varying the ω angle. Figure 6(a) shows the observed box-integrated intensities, $\mathcal{I}^{++}_{\mathrm{obs}}(\omega_n, t_{2n-1})$ and $\mathcal{I}^{+-}_{\mathrm{obs}}(\omega_n, t_{2n})$. The polarization correction was not applied to these data. A finite spin-flip intensity was observed owing to the imperfect neutron polarization. Then, we applied the polarization correction to all the observed intensity maps and converted them into an intensity distribution in reciprocal space. Figure 6(b) shows the profiles of the corrected non-spin-flip and spin-flip intensities, I^{++} and I^{+-}, along the $(H, 0, 2)$ line, which were sliced from the three-dimensional data. Since only the non-spin-flip scattering is expected for nuclear scattering, this result indicates that the polarization correction was successfully performed. Details of the polarization correction are described in Appendix A.

B. Polarization analysis of magnetic scattering

We applied the polarization analysis with the ^{3}He-NSF and the 2D detector to the study of magnetic structures of Mn_3WO_6. As mentioned in the introduction, this compound displays two antiferromagnetically ordered phases. In this paper, we focus on the high-temperature phase and show the data measured at 23 K. First, we determined the q-vector of the magnetic order by unpolarized neutron diffraction measurements with the 2D detector. The horizontal scattering plane was set to the $(H, 0, L)$ plane with the c axis perpendicular to the incident neutron beam. We measured intensity maps by scanning the ω angle from $-52.9°$ to $-46.0°$ in steps of $0.1°$ and converted the resulting data into reciprocal-space coordinates. Figure 7(d) shows the intensity distribution projected onto the $(H, 0, L)$ plane. We observed two incommensurate reflections near $(-1, 0, 1.5)$. To investigate the position of the reflections in more detail, we plotted the intensity distribution on the $(H, K, 1.5)$ plane, which is perpendicular to the horizontal plane, as shown in Fig. 7(b). This reveals three incommensurate peaks surrounding $(-1, 0, 1.5)$. One of the reflections (labeled q_1 in Fig. 7(b)) can be indexed using the q-vector $\boldsymbol{q}_1 = (0.059, 0.019, 0.5)$. The other two reflections can be indexed using $\boldsymbol{q}_2$ and $\boldsymbol{q}_3$, which are obtained by threefold rotations of $\boldsymbol{q}_1$ about the c^* axis. This relationship is consistent with the threefold rotational symmetry of the crystal structure about the c axis. Note that the unequal distribution of the magnetic intensities would be due to the orientations of the Fourier-transformed magnetic moments with respect to the direction of the scattering vector. We emphasize here that $\boldsymbol{q}_1$, $\boldsymbol{q}_2$, and $\boldsymbol{q}_3$ cannot lie within a single 2D plane passing through the reciprocal-space origin as shown in Fig. 7(a). This highlights the importance of three-dimensional reciprocal-space measurements for elucidating such complex incommensurate magnetic structures.

We thus measured the spin-flip and non-spin-flip intensity distributions around $(-1, 0, 1.5)$ using the ^{3}He-NSF. Figures 7(c) and 7(e) show the intensity difference $I^{++} - I^{+-}$ mapped onto the $(H, K, 1.5)$ and $(H, 0, L)$ planes, respectively. We clearly observed that the difference in intensity takes positive and negative values for the reflections $\boldsymbol{q}_1$ and $\boldsymbol{q}_3$, respectively. To compare the spin-dependent intensities more quantitatively, line profiles along the H direction were obtained by projecting the intensities within the dashed-box regions corresponding to $\boldsymbol{q}_3$ and $\boldsymbol{q}_1$ in Fig. 7(c). The resulting line profiles are shown in Figs. 7(f) and 7(g), respectively. At the reflection corresponding to $\boldsymbol{q}_3$, we found that the magnetic scattering intensity was dominated by spin-flip scattering, while at $\boldsymbol{q}_1$, the non-spin-flip scattering was more prominent. The asymmetry of the line profiles, defined as $(I^{++} - I^{+-})/(I^{++} + I^{+-})$, was calculated from the integrated intensities, and the results are summarized in Table II. Since the neutron polarization is parallel to the z axis defined in Fig. 7(a), the spin-flip scattering arises from the Fourier-transformed magnetization components in the xy plane, which contains the a and c axes. If the magnetic moments have only a c-axis component, as illustrated in Fig. 8(a), the calculated asymmetry is -0.99 for both reflections. In this case, both reflections would appear predominantly in the spin-flip channel, which is inconsistent with the experimental results. We therefore considered an oblique sinusoidal model, as illustrated in Fig. 8(b). The moment direction is specified by the polar angle θ, measured from the c axis, and the azimuthal angle ϕ, measured from the a axis. A least-squares analysis was performed to determine θ and ϕ, yielding $\theta = 49.5°$ and $\phi = 108.5°$. A more detailed structural analysis, including data from other regions of reciprocal space, will be reported elsewhere [25]. The present results demonstrated that polarized neutron scattering with a ^{3}He-NSF and a 2D detector is quite useful for solving complex incommensurate magnetic structures.

TABLE II. Measured and calculated asymmetries for the $(-1, 0, 2) - \boldsymbol{q}_1$ and $(-1, 0, 2) - \boldsymbol{q}_3$ reflections. The asymmetry is defined as $(I^{++} - I^{+-})/(I^{++} + I^{+-})$. The experimental values were obtained by integrating the line profiles shown in Figs. 7(f) and 7(g). The c-axis-parallel and oblique sinusoidal models of the magnetic moment $\mathbf{M}$ are illustrated in Figs. 8(a) and 8(b), respectively.

	$(-1, 0, 2) - \boldsymbol{q}_1$	$(-1, 0, 2) - \boldsymbol{q}_3$
Experiment values	0.30 ± 0.04	-0.49 ± 0.10
$\mathbf{M} \parallel c$ sinusoidal model ($\theta = 0°,\ \phi = 0°$)	-0.99	-0.99
Oblique sinusoidal model ($\theta = 49.5°,\ \phi = 108.5°$)	0.30	-0.50

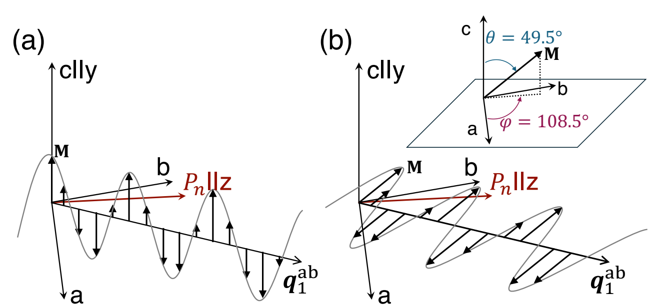

FIG. 8. Schematic illustrations of the sinusoidal magnetic structures. (a) the c-axis-parallel sinusoidal model and (b) the oblique sinusoidal model. The $\boldsymbol{q}_1^{\mathrm{ab}}$ vector is the projection of the magnetic propagation vector $\boldsymbol{q}_1$ onto the ab plane. The P_n vector is the neutron polarization direction, which is parallel to the z axis in Fig. 7(a). Inset of the panel (b) shows the definition of the polar angle θ and azimuthal angle ϕ. In the oblique sinusoidal model, the magnetic moment direction $\mathbf{M}$ is specified by $\theta = 49.5°$ and $\phi = 108.5°$.

V. SUMMARY AND OUTLOOK

In this work, we demonstrated polarized neutron diffraction measurements combining the ^{3}He-NSF and the 2D detector to explore three-dimensional reciprocal space. We established the correction procedure for the time-dependent neutron polarization and transmission using a nuclear Bragg reflection and the NMR signal, and confirmed its validity. The polarization analysis for the incommensurate magnetic reflections of Mn_3WO_6 was successfully performed, highlighting its importance for magnetic reflections outside the horizontal plane. Although the present *ex-situ* method requires time-dependent correction, it is possible to upgrade it to an *in-situ* SEOP method. In fact, a compact *in-situ* ^{3}He-NSF system has been recently developed at J-PARC [38]. This implementation is expected to maintain high neutron polarization and transmission throughout the experiment, thereby facilitating polarization analysis of complex magnetic structures over a wider region of reciprocal space and for a broader range of materials.

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Appendix A: Polarization correction with an *ex-situ* ^{3}He neutron spin filter

In this section, the polarization correction in the present experimental setup is described. This correction is based on the polarization matrix reported by A. Wildes [39, 40]. A count obtained when both the polarizer and the analyzer are set to the spin-up state, I_{obs}^{++}, is expressed as the sum of the contributions from the four scattering channels:

$$\begin{aligned} I_{\mathrm{obs}}^{++} &= N^+ 2T_p p_p I^{++} 2T_a p_a \\ &+ N^+ 2T_p p_p I^{+-} 2T_a (1-p_a) \\ &+ N^- 2T_p (1-p_p) I^{-+} 2T_a p_a \\ &+ N^- 2T_p (1-p_p) I^{--} 2T_a (1-p_a). \end{aligned} \tag{A1}$$

Here, N^+ and N^- denote the numbers of up and down neutrons incident on the polarizer from the neutron source, respectively. T_p and T_a are the unpolarized neutron transmissions of the polarizer and analyzer, respectively. The polarization efficiencies of the polarizer and analyzer are defined as $p_p = (1+P_p)/2$ and $p_a = (1+P_a)/2$, respectively, where P_p and P_a are the neutron polarizations produced by the polarizer and analyzer. These parameters take values in the ranges $0 \le T_p, T_a, P_p, P_a \le 1$, and $0.5 \le p_p, p_a \le 1$. I^{++}, I^{+-}, I^{-+}, and I^{--} are the scattering intensities for the corresponding spin states. The derivation of this equation is schematically illustrated in Fig. 9. In practice, this equation also includes a background term, which is ignored here for simplicity.

In the present experimental setup, only the I_{obs}^{++} and I_{obs}^{+-} configurations can be measured. If the ^{3}He-NSF analyzer is switched to the spin-down state, the factors for the up and down neutrons, $2T_a p_a$ and $2T_a(1-p_a)$, are interchanged. Here, we define the total number of incident neutrons as $N_0 = N^+ + N^-$. Assuming an unpolarized incident beam coming from the neutron source, we have $N^+ = N^- = N_0/2$. N_0 is usually normalized by the counting time or monitor counts. As also noted in Sec. III, the sample is an antiferromagnet and its nuclear and magnetic reflections are separated from each other. Moreover, the chiral magnetic scattering, which depends on the sign of the incident neutron polarization direction, can be neglected because the neutron polarization is nearly perpendicular to the scattering vector. With these conditions, $I^{++} = I^{--}$ and $I^{+-} = I^{-+}$ hold, allowing Eq. A1 to be simplified into the following matrix form in terms of the non-spin-flip and spin-flip intensities:

$$\begin{bmatrix} I_{\mathrm{obs}}^{++} \\ I_{\mathrm{obs}}^{+-} \end{bmatrix} = 2T_p T_a \begin{bmatrix} p_p & 1-p_p \\ 1-p_p & p_p \end{bmatrix} \begin{bmatrix} p_a & 1-p_a \\ 1-p_a & p_a \end{bmatrix} \begin{bmatrix} I^{++} \\ I^{+-} \end{bmatrix}. \tag{A2}$$

Since the ^{3}He polarization of the *ex-situ* ^{3}He-NSF analyzer decays over time, both the neutron polarization efficiency p_a and the neutron transmission T_a become time-dependent. Accordingly, by treating Eq. A2 as a time-dependent matrix equation and taking its inverse, the following expression is obtained:

$$\begin{bmatrix} I^{++} \\ I^{+-} \end{bmatrix} = \frac{1}{2T_p P_p \left(P_a(t_1) + P_a(t_2)\right)} \begin{bmatrix} 1 + P_p P_a(t_2) & -1 + P_p P_a(t_1) \\ -1 + P_p P_a(t_2) & 1 + P_p P_a(t_1) \end{bmatrix} \begin{bmatrix} 1/T_a(t_1) & 0 \\ 0 & 1/T_a(t_2) \end{bmatrix} \begin{bmatrix} I_{\mathrm{obs}}^{++}(t_1) \\ I_{\mathrm{obs}}^{+-}(t_2) \end{bmatrix}, \tag{A3}$$

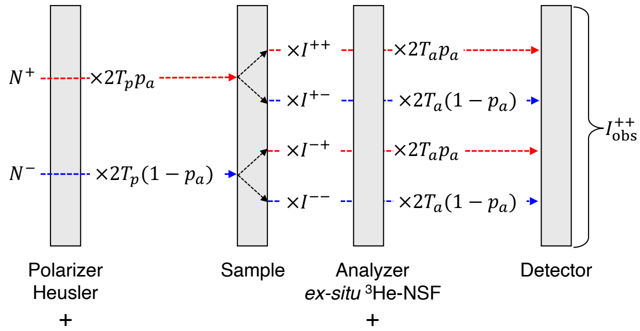

FIG. 9. Schematic diagram of the spin-dependent neutron intensities in the present experimental setup when the Heusler polarizer and the ^{3}He-NSF are both set to select up-spin (+) neutrons. The red and blue lines indicate the paths of the up neutrons (N^+) and down neutrons (N^-), respectively. The observed intensity, I_{obs}^{++}, is obtained as the sum of the contributions from the four spin-scattering channels.

where p_p and p_a are converted to P_p and P_a using the relations $P_p = 2p_p - 1$ and $P_a = 2p_a - 1$. $T_a(t)$, P_p, and $P_a(t)$ are determined experimentally, as described in Sec. IV A. In contrast, T_p of the Heusler polarizer does not depend on time and is treated as a single constant scale factor. This time-dependent correction was applied to each detector pixel, allowing data collected throughout the entire measurement period to be used for the polarization analysis.

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